

A course in language creation



Conlangs University

Morphology 1

Syllables & words



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Moving up the food chain from phono, now it's time for morpho. First we'll talk about different types of morphology and how linguists write morphemes. Then we'll talk about different tendencies languages tend to have with respect to how their morphemes fit together (or don't!) Last, we'll talk about different forms morphemes can take. These are grouped under **concatenative** in which morphemes are strung together linearly and **nonconcatenative** in which they may be discontinuous.

Types of morphology

Inflection vs derivation

Morphology can be split into inflection and derivation. **Inflectional morphology** consists of morphemes added to a word to mark it for different categories or functions within the context of a sentence. It doesn't change the base meaning of the word, but situates it within a phrase. Examples in English are suffixes like *-s* which marks a noun as plural and *-ing* which marks the gerund form of a verb. Inflectional morphology does not change the part of speech and inflected forms tend to be considered "the same word." Inflection of verbs can also be called **conjugation** and inflection of nouns and adjectives can also be called **declension**.

Derivational morphology is a process that takes morphemes and combines them to form new words. Often these are affixes that take words and create new words with predictably related meanings. For example in English, *un-* takes an adjective and derives another adjective with the opposite meaning and *-ness* takes an adjective *x* and derives a noun meaning "the condition of being *x*" or "the degree to which something is *x*." Derivation can change a word's part of speech, and forms related by derivation tend to be considered different words. Another form that derivation can take is **compounding** where two stems are combined into a new word whose meaning relates to both.

Productivity

A morphological pattern is said to be **productive** if it can apply freely to new words in the relevant class. The English plural morpheme *-s* is productive because you can grammatically apply it to any new noun you encounter. This can work with almost all native words in English, with words English loans from other languages (one emoji, two emojis), and even with words you've never encountered before. If I handed you a *sprongle*, and then I handed you another one, you'd know that you had two *sprongles*.

Morphology is **non-productive** on the other hand if it cannot apply to new words. A non-productive morpheme has a defined meaning and exists in certain words, but it can't be applied to just any word. They can be derivational or inflectional. The plural suffix *-en* is an example of a non-productive inflectional suffix. It marks the category of plurality in words like **children**, **oxen**, and **brethren**, but can't be applied to new words in the same way its productive counterpart *-s* can. One *wug* doesn't become two *wuggen*. An example of non-productive derivational morphology in English is the suffix *-(a)tion*, which derives an action or result noun from a related verb. For example from *construct* you get *construction* and from *adapt* you get *adaptation*. We have an intuitive understanding about what that suffix means, but we can't add it on to any word. "*Helpation*" doesn't mean "the act or result of being helped."

Glossing

In order to give information about how a language's morphology breaks down to people who aren't familiar with the language, linguists developed a tool called **glossing**. When you gloss a sentence, you break it down into its constituent parts and write what they all mean separately. Here's an example of a glossed sentence in Spanish.

1. Muchas personas escucharan la radio.
much-as person-as escuch-aran la radio
many-F.PL person-F.PL listen-FUT.3.PL DEF.F.SG radio
"Many people will listen to the radio."

First you give the sentence in the source language, sometimes as it appears originally, followed by a version that's broken down into morphemes. Then, underneath that you give a "gloss" for each morpheme. Roots of content words are glossed with loose translations, such as *escuch-* **listen**, *person-* **person** and *radio* **radio** (I picked this example cause it turns out English and Spanish are closely related. Who knew.) Morphemes that are part of the same word, but separable are linked using a hyphen. When a morpheme has multiple meanings but they can't be separated, the meanings are linked using a period. Morphemes that convey grammatical information are glossed using standardized abbreviations for the grammatical categories they mark, for example *-as* **F.PL** marks **feminine plural** on adjectives and nouns, and *-aran* **FUT.3.PL** marks **future tense** with a 3rd person **plural** subject.

I'm going to use glossed example sentences to show each of the types of morphology below. I'll explain anything new that comes up and help get you used to glosses, which you'll end up seeing all the time around here. If you want a thorough explanation and reference, check out the [Leipzig Glossing Rules](#).

Morphological typology

Morphological typology is a way of describing the general tendencies a language has when combining elements to form words and build meaning. The smallest division of a language which has meaning is called a **morpheme**. Some morphemes can occur by themselves in a sentence. These are called **free morphemes**. Examples from English include lexical words like "cat" and function words like "and". Other morphemes cannot occur alone and have to be attached to some other morpheme. These are called **bound morphemes** and examples from English include "-s" and "-ing". Often things that speakers consider to constitute a single word, like "cats" can be broken down into multiple morphemes; here we have "cat" indicating a small feline and "-s" indicating that there are more than one of them. Languages have different tendencies when combining morphemes. Although no language entirely follows one typology entirely, they can be an interesting and useful lens to look at conlangs through. Morphological typology can loosely be viewed on two axes, which this chapter will outline.

Analytic vs synthetic

The first axis contrasts analytic languages and synthetic languages. An **analytic** language has a low number of morphemes per word, whereas a **synthetic** language has a high number of morphemes per word. It's important to remember that these are relative and general tendencies, rather than absolute truth about a language. It's common for languages to have some more analytic constructions and some more synthetic constructions.

The most canonically analytic languages are found in East Asia and West Africa. English is also fairly analytic. Analytic languages often have particles, auxiliary verbs, or other words that serve grammatical functions, rather than relying on prefixes and suffixes. Take this sentence from the analytic language Cantonese (Sinitic, Sino-Tibetan).

```
1. lei5 ge3 pang4jau5 wui6 hai2 heung1gong2 jyu2
2 POSS friend FUT in Hong Kong live
   "Your friend will live in Hong Kong."
```

Each word in this sentence is just one morpheme. Grammatical categories like tense are marked with particles. Relations like possession and location are marked using standalone particles. Languages that take this to the extreme, and consist of *only* free morphemes are called **isolating**.

Compare the analytic sentence above with a similar sentence in a typical synthetic language, Turkish (Turkic). Synthetic languages tend to express grammatical categories and relations using inflectional affixes, which are bound morphemes.

2. sen-in arkadaş-in İstanbul-da otur-acak
 2sg-GEN friend -2sg Istanbul-LOC live-FUT
 "Your friend will live in Istanbul."

In this example, the relations of possession and location are marked using bound morphemes, the object of possession is marked using an affix showing its possessor, and the tense is marked on the verb with an affix. Each word consists of multiple morphemes, many of which do not occur individually. Where analytic languages have series of free morphemes that combine to give meaning, synthetic languages tend to combine many bound morphemes into longer words to give meaning. In extreme cases, this can result in long words that would correspond to whole phrases in more analytic languages. Take these examples from Tuscarora (Iroquoian) and Guaraní (Tupian) respectively, in which a single word carries the same meaning that takes a sentence to express in English.

3. n- a- kə- taʔn-yə- ʔθe- hr- aʔ
 CISLOC-FAC-1SG/2SG-camp-enter-APPL-AND-P
 "I should come visit you."
4. nd- a- guata-se- ve- i- ma
 NEG-1SG-walk- DES-AUG-NEG-PFV
 "I already do not want to walk anymore."

Languages with this sort of tendency are often called **polysynthetic**, since they are synthetic languages that combine many parts. The boundaries of polysynthesis are fairly vague, so it can be difficult to say for sure what does or doesn't count.

Fusional vs agglutinating

The second axis of morphological typology contrasts fusional and agglutinating languages. A **fusional** language is one where a single morpheme carries multiple meanings. European languages are often fusional, for example Spanish (Romance, Indo-European).

5. much-as person-as escuch-aran la radio
 many-F.PL person-F.PL listen-FUT.3.PL DEF.F.SG radio
 "Many people will listen to the radio."

The ending *-as* marks both the noun *personas* and its modifier as being feminine plural. The verb ending *-aran* marks the verb as future tense, with a plural, third-person subject. The determiner *la* marks the noun as definite, feminine, and singular. The number of categories marked in a single morpheme is called its **exponence**. Each one of these inflectional affixes codes for more than one category. Inflectional affixes in fusional languages tend to have higher exponence.

Agglutinative languages, on the other hand, tend to use a higher number of morphemes with lower exponence each.

6. arkadaş-lar-ım ev -im -e gid-iyor-muş-lar
friend -PL -1sg house-1sg-ALL go -PROG-EVD-PL
"My friends [must] have been going to my house."

In this example it's clear that in Turkish, number, possessor, and case all have distinct affixes that are stacked to convey category. Similarly, the verb is marked for aspect, evidentiality, and agreement with the plural subject using separate, distinct affixes. This contrasts with the multi-exponent affixes you tend to see in more fusional languages.

Concatenativity

A piece of morphology is considered **concatenative** if it involves stringing together parts in an order, where parts may influence each other, but still maintain a clear order.

Prefixes, suffixes, circumfixes

You're probably already familiar with prefixes and suffixes, since we use them widely in English. **Prefixes** are morphemes that come before a root, for example English *un-*, which takes an adjective and derived another one that means the opposite, for example *happy* vs. *unhappy*. **Suffixes** are ones that come after the root, such as the English plural marker *-s*.

Circumfixes are morphemes that consist of two parts: one that comes before the root and one that comes after. Sometimes one part or the other can be used separately, but they must have a unique meaning when used together to justify being considered a single morpheme. A common example of a circumfix is the German past participle marker *ge-*.

```
2.  ich habe das  schon  ge -mach-t  
    I   have that already PST-do  -PST  
    "I have already done that."
```

Usually you gloss both parts of circumfixes, but sometimes people will gloss one part with **CIRC** to indicate that it's part of a circumfix rather than a second morpheme. So you might see *gemacht* glossed as **PST-do-CIRC** instead.

Another thing to note in this gloss is that you don't always have to break words down into their constituent morphemes when it's not relevant. I could translate *ich* as "I" or I could break it down as a first person singular nominative pronoun and write **1s.NOM**. If it's relevant to the example, it's always better to break it down, but if it doesn't matter, it doesn't matter.

Infixed

Infixed are affixes that get placed *inside* the root, breaking it up. English does this in a marginal but kinda famous way. We don't have any inflection or derivation that can go inside morphemes, but we do have swear words! English allows you to put swear words like *fucking*, or *bloody* for any Brits out there, in the middle of words to intensify them. You might say my use of swear words in an intro course is *irre-fucking-sponsible*, to which I say, it *abso-fucking-lutely* is not. It turns out it's not just random: there's a grammatical rule dictating where expletives can be stuck into words. Expletives have to be put in at the syllable break

right before the syllable with the primary stress. Think about it. *Ab-fucking-solutely* and *absolute-fucking-ly* both sound wrong, right?

Some languages use infixes for inflection and derivation. Many Austronesian languages, such as Tagalog, have verb conjugations that manifest as infixes after the first consonant of a word. Infixes are glossed in angle brackets.

```
3. Kumain ka na.  
   <um>kain ka na  
   <PRF>eat you already  
   "You have already eaten."
```

The Tagalog perfect tense is marked using an infix *<um>* which is inserted after the first consonant of the verb. What happens if there's no consonant to insert it after?

```
4. Umalis ang lahat.  
   <um> alis ang lahat  
   <PRF>leave ART everyone  
   "Everyone left."
```

It shows up as a prefix. It's extremely rare to have infixes that don't show up as prefixes or suffixes somewhere. It's often considered that infixes like these occur because of prefixes or suffixes that create consonant clusters that don't meet the language's phonotactics. For example **umkain* contains /mk/ which is not allowed in Tagalog. (A * before a word or phrase indicates that it's ungrammatical or not allowed for some reason.) So your options are to change the prefix or to swap the sounds around and get /kumain/.

A note on clitics

Clitics are morphemes that fall somewhere between words and affixes. Words can exist independently but affixes must be stuck on to another word. Affixes also mark meaning on a particular word, but clitics may attach to multiple word types and often mark meaning on a phrase or sentence level. The boundaries between words, clitics, and affixes are fuzzy and often depend on language-specific criteria.

A familiar example of a clitic is the English possessive *'s*. Note that rather than with a hyphen, clitics are connected to their host with an equals sign. I've glossed *'s* here as **POSS** for possessive and *the* as **DEF** for definite.

5. I went to [my mother]=s house
1s.NOM go.PST to [1s.POSS mother]=POSS house
"I went to my mother's house."
6. I saw [the queen of england]=s palace
1s.NOM see.PST [DEF queen of england]=POSS palace
"I saw the Queen of England's palace."
7. I found [the man with the yellow hat]=s monkey
1s.NOM find.PST [DEF man with DEF yellow hat]=POSS monkey
"I found the man with the yellow hat's monkey"

Pay attention to where the -'s attaches. If it were a regular affix you'd expect it to attach to the head of the phrase--the word that most closely passes its meaning to the meaning of the phrase, in this case "mother", "queen," and "man." But it doesn't! It attaches to the last word in each phrase and it modifies the meaning of the whole phrase. It's placed as though it were its own word, but pronounced as though it were part of the previous word.

Non-concatenativity

Nonconcatenative morphology consists of processes where some of the morphemes are not easily separable from others, either because they take their form from the root they're modifying or because they manifest by changing something about the root rather than adding something on.

Reduplication

Reduplication is a process where part or all of a word is copied as part of a morphological process. It often serves an iconic purpose to indicate a greater degree of something, such as pluralization of nouns, or marking iteration or continuation of verbs. It can be total, where the whole word is repeated, or partial, where just part of the word is repeated. The Motu language from New Guinea uses both: **mahuta** means "to sleep." Reduplicating the whole word gives you **mahutamahuta** "to sleep constantly," while reduplicating the first syllable gives you **mamahuta** "they sleep." Reduplication is glossed using a tilde and the abbreviation **REDUP** or **RDP**, such as in this example from Indonesian (the demonstrative *itu* is glossed as **DEM**).

```
8. Siapa orang-orang itu?
   siapa orang ~REDUP itu
   who   person~PL   DEM
   "Who are those people?"
```

Apophony

Apophony refers to morphological operations realized by changing some sound in the original word in a predictable way. English has a couple of good examples of this too. **Umlaut** is an instance of apophony found in Germanic languages involving using back/front vowel alternations to indicate singular/plural forms of nouns, like *foot/feet* and *goose/geese* or *mouse/mice* and *louse/lice*. This dates back to an earlier plural suffix *-i*. The last vowel in a word with *-i* tended to assimilate/harmonize and become more front. The suffix was lost and we ended up with pairs that only different in vowel frontness. Incidentally this helps explain the age-old question why the plural of *moose* isn't *meese*. The word "moose" entered the English language from an Algonquian language long after the *-i* suffix had disappeared, so it never underwent that change.

Another case of apophony is **ablaut** which involves grammatical vowel alternation in Indo-European languages. This was inherited into English as its many series of irregular verbs which are conjugated by changing their vowels, for example *sing/sang/sung* or *drive/drove/driven*. Apophony is glossed using a backslash.

```
9. the geese flew
   DEF goose\PL fly\PST
   "The geese flew."
```

Apophony can also affect consonants, for example English has a fair number of verb/noun pairs which differ only in that the verb has a voiced final consonant and the noun has an unvoiced final consonant: *to believe* vs. *a belief*, *to house* vs. *a house*, *to use* vs. *a use* and so on.

Consonant mutation

Celtic languages are well-known for their systems of consonant apophony, where initial consonants change in predictable ways depending on various grammatical constructions. These are called **consonant mutations**. Let's take a look at how these came about in Irish and then see some examples.

Imagine you have the words /in/ "in," /as/ "her," /an/ "their," /en/ "bird" and /tjax/ "house." I'm not gonna bother with Irish orthography here, so hopefully it'll be clearer. In the earliest form of the language, you can just combine them linearly like /an en/ "their bird," /as tjax/ "her house," /in tjax/ "in a house," /an tjax/ "their house" or /in an tjax/ "in their house." Then sound changes start to hit. First, unvoiced stops get voiced after nasals, so /an tjax/ is realized as [an djax] and /in an tjax/ is realized as [in an djax]. Since there's no nasal to cause the sound change, "her house" /as tjax/ keeps the voiceless stop [as tjax]. Next, nasals disappear except between vowels. So now the original /an tjax/ becomes [a djax] and /in an tjax/ becomes [in a djax]. Next suppose that /s/ becomes [h] intervocalically (between vowels) and is lost otherwise. Now the words for "her" and "their" sound the same. The only thing that's left to differentiate between them is whether or not the sound change occurred in the following word. So "in her house" /in as tjax/ is [in a tjax] and "in their house" /in an tjax/ is [in a djax].

It's also interesting to see what happened to phrases with the word /en/, which starts with a vowel. "Her bird" /as en/ becomes [ah en] and "their bird" /an en/ is [an en]. With time, the original phonemic analyses don't really make sense, because you don't see an /s/ or /n/ in surface forms and both pronouns look more like /a/, just with different effects on the following word. So the leftover consonant before vowels is treated like part of the mutation. That's why you end up with /h/ getting stuck on the beginning of certain words in some contexts and /n/ getting stuck on the beginning in other contexts. That also explains the apparently irregular combining forms of the preposition *i*. Here's a table that summarizes the changes.

Original	Voicing after Nasals	Loss of Nasals	Loss of S	Reanalysis
in tjax	in djax	i djax	i djax	i djax
an tjax	an djax	a djax	a djax	a djax
in an tjax	in an djax	in a djax	in a djax	ina djax
as tjax	as tjax	as tjax	a tjax	a tjax
in as tjax	in as tjax	in as tjax	in a tjax	ina tjax
as en	as en	as en	ah en	a h-en
an en	an en	an en	an en	a n-en

Now you can see how this kind of thing happens over time, which should shed some light on how to design a system like it if you choose to use one in your conlang.

Disfixation

Disfixation is the process of marking something morphologically not by adding something to a word, but by removing part of one. They're rare but they are attested in some Native American languages as well as possibly in French. French has a phenomenon where word-final consonants and word-final *e* are dropped. Historically, to form feminine forms of adjectives, you'd add an *-e* on to the end, but nowadays the only effect that *-e* has is to keep the final consonant from being deleted. This has created sets of paradigms where the feminine form is not predictable from the masculine form, but the masculine form is usually made by removing the last consonant of the feminine form. *Small* and *white* are /pətit/ and /blāʃ/ when feminine and /pəti/ and /blā/ when masculine. If you treat the feminine form as the unmarked one, then the masculine looks like it's marked with a disfix!

Consonantal roots

Semitic languages have an interesting and unusual morphological pattern where the base meaning of a word is carried by a three-consonant root, and inflection and derivation are shown by putting those three consonants in a variety of templates. In Arabic and Hebrew, the three-letter root K-T-B denotes things having to do with writing. If you put it in the present-tense verb template CaCaCa, you get *kataba* "he writes". If you put it in the object noun template CiCaC, you get *kitab* "a book, that which is written". If you put it in the location template maCCaC, then you get *maktab* "office, a place where you write".

When evolving consonantal roots, think about how other types of non-concatenative morphology evolve. Suppose you have a word like /kopob/. Pluralize by adding an -i, and the second to last vowel gets fronted (like umlaut) and you get /kopøbi/. Lose word-final vowels and you get /kopøb/. Morpho through only a vowel shift. Now suppose you have word-initial stress and mark genitive with a prefix a-. Stressed vowels get lengthened so you contrast /'ko:pob/ with /'akopob/. Then long /o:/ becomes /u/ and word-initial /a/ get lost. Now you contrast /kopob/ and /kupob/ in the singular and /kopøb/ and /kupøb/ in the plural. Now suppose you mark accusative with a prefix he-. You'd get /'hekopob/. Reduce the vowel after a stressed syllable and now you contrast /kopob/ and /hekpob/ in the singular and /kopøb/ and /hekpøb/ in the plural. Just from concatenative morphology and sound changes, you've started to get something that looks an awful lot like a root k-p-b with templates like CoCoC, CoCøC, CuCoC, heCCoC.

Conclusion

Here I've gone over the differences between derivational and inflectional morphology as well as between productive and non-productive morphology. I've given an introduction to glossing and several examples over the course of the reading. Don't forget to check out the [Leipzig Glossing Rules](#)! Then I explained the tendencies that languages can follow when putting their parts together. Next I talked about the different forms affixes can take and how several types of well-known nonconcatenative morphology came to be. Think about how it applies to your conlang! Does your conlang tend to be more analytic or synthetic? Do different groups of words behave differently? What kinds of morphology are common for you?